Materials Science - Project 3
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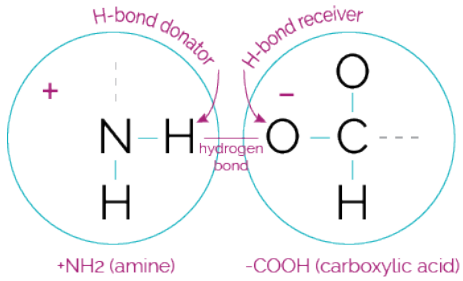
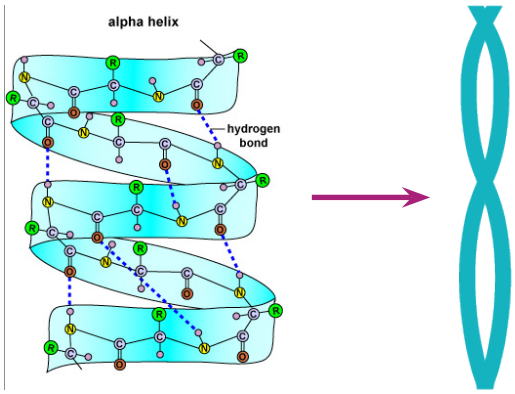
# D4M4GED

## Exploring the material effects of bleach on hair

### WHAT IS HAIR?

Hair is mostly made out of the fibrous protein alpha-keratin. The primary structure of keratin in hair is formed by peptide and hydrogen bonds.



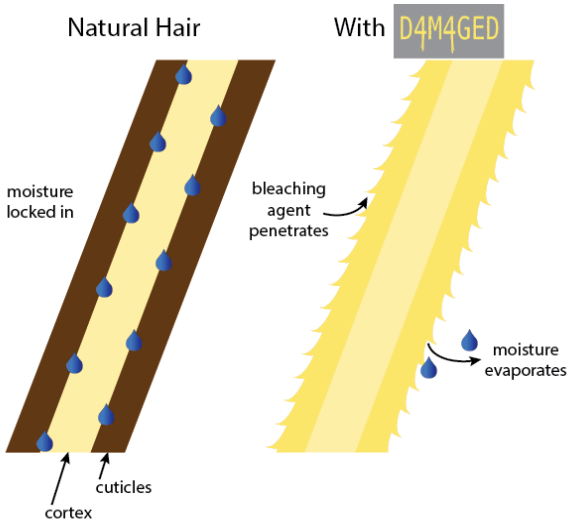

The secondary structure of keratin in hair is composed of alpha helices. Alpha helices involve long polypeptide chains of the proteins above which are twisted into helices by hydrogen bonds. Hydrogen bonds exist between the hydrogen (H-bond donor) and oxygen (H-bond receiver) atoms along each chain. An anti-parallel pair of two types of alpha helices form rope-like structures. When hair breaks, secondary bonds which stabilize the helix structure fail and cause the chains to fail and unravel.

The natural color of hair comes from two types of melanin protein - eumelanin and phaeomelanin. Everybody has a ratio of eumelanin and phaeomelanin. A larger eumelanin content causes hair to be brown or black, while more phaeomelanin causes hair to be red or blond. When melanin is no longer present, this causes hair to turn white or grey.

### WHAT HAPPENS WHEN HAIR IS BLEACHED?

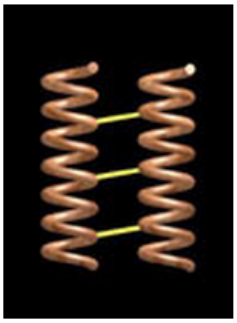
Bleach oxidizes the melanin molecules, which causes them to become colorless. leaving behind the natural yellow color of keratin.

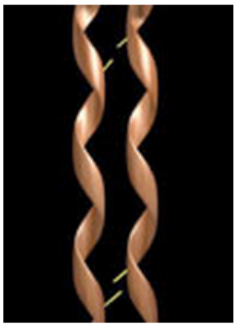
During the bleaching process, the oxidizing agent - often peroxide - lifts the cuticles (flakes) of the hair in order to fully penetrate the cortex. With excessive bleaching, however, the cuticles become permanently lifted, allowing for moisture to continuously evaporate from the hair and make the hair brittle. These lifted cuticles reflect light in a way that makes the hair less shiny.

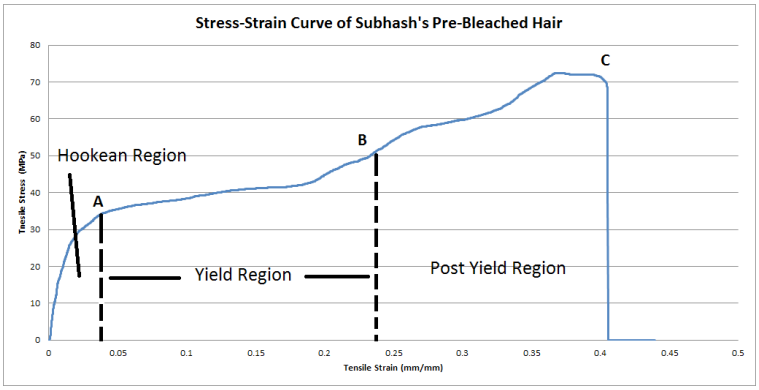


## MECHANICAL PROPERTIES OF HAIR

Due to the fact that microfibrils and and the matrix are parallel one another, stretching requires simultaneous deformation of these structural parts. During the beginning of the stretching process to before the hair breaks, there are two processes that happen. Before breaking, hair undergoes an elastic behavior like a rubber band. Hair can stretch up to 20-50% of its original length before breaking, depending on the speed of the force pulling the hair and the wetness of hair.



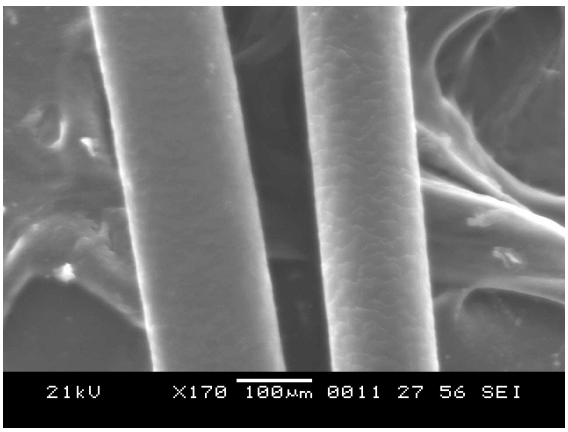


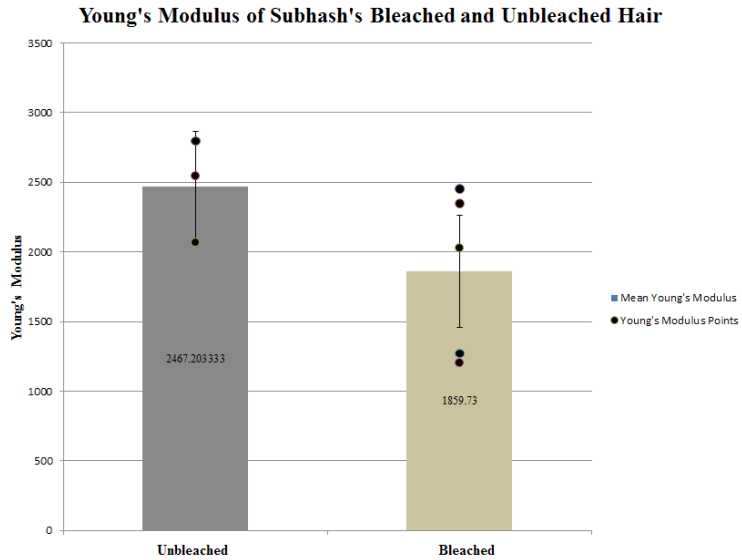



The Hookean region is the approximately linear phase, where the hair has mechanical characteristics of a purely Hookean spring in its properties. During the Yield region, the hair continues to extend and the stress of the hair does not change greatly. For extensions beyond the Yield region, the fiber becomes stiffer with extension and further strain before there is a force at the end of the Post-Yield Region that breaks the hair at a percentage of extension.

## OUR TESTING

Under the Scanning Electron Microscope (SEM), we placed a bleached and unbleached sample side by side to compare. As you can see, the bleached hair (right) has lifted cuticles, with much more jagged edges compared to those of the unbleached hair. These raised cuticles lead to a loss of sheen and additional frizziness.




To determine some of the mechanical properties of our hair, we decided to test individual strands under a tensile test. We placed ten samples of each type of hair, bleached and unbleached, between two pieces of plastic using super glue. A successful test occurred when the hair snapped somewhere in the middle of the sample, not slipping or breaking at a stress concentration. Unfortunately, out of our ten samples, only three unbleached and five bleached were successful tests. Due to time constraints, we were not able to perform the test with more samples and get more definitive data.